

Assignment No. 6: Data Manipulation and Table Management

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Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
City	VARCHAR(30)	
CourseID	INT	FOREIGN KEY

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Course (
```

```
    CourseID INT PRIMARY KEY,
```

```
    CourseName VARCHAR(50) NOT NULL,
```

```
    Fees DECIMAL(10,2) NOT NULL
```

```
);
```

CREATE TABLE Student (

StudentID INT PRIMARY KEY,

StudentName VARCHAR(50) NOT NULL,

Age INT,

City VARCHAR(30),

CourseID INT,

FOREIGN KEY (CourseID) REFERENCES Course(CourseID)

);

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. A SQL query has been executed, resulting in 10 rows being inserted into the 'StudentResult' table. The query is displayed in the SQL tab, and the console shows the execution details.

```
INSERT INTO StudentResult (StudentID, StudentName, Course, City, Marks, FeesPaid) VALUES (1,'Rahul Sharma','BSc IT','Mumbai',78,45000), (2,'Priya Patil','BCA','Pune',85,40000), (3,'Aman Verma','BSc CS','Nashik',67,42000), (4,'Sneha Joshi','BSc IT','Mumbai',91,45000), (5,'Karan Mehta','BCA','Thane',74,40000), (6,'Neha Shah','BSc IT','Pune',88,45000), (7,'Rohan Desai','BSc CS','Mumbai',69,42000), (8,'Meera Kulkarni','BCA','Nashik',82,40000), (9,'Arjun Nair','BSc IT','Thane',59,45000), (10,'Divya More','BCA','Mumbai',93,40000);
```

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The 'StudentResult' table is displayed in the 'Browse' tab. The table contains 10 rows of data, including StudentID, StudentName, Course, City, Marks, and FeesPaid. The interface includes various options for editing, deleting, and exporting the data.

StudentID	StudentName	Course	City	Marks	FeesPaid
1	Rahul Sharma	BSc IT	Mumbai	78	45000.00
2	Priya Patil	BCA	Pune	85	40000.00
3	Aman Verma	BSc CS	Nashik	67	42000.00
4	Sneha Joshi	BSc IT	Mumbai	91	45000.00
5	Karan Mehta	BCA	Thane	74	40000.00
6	Neha Shah	BSc IT	Pune	88	45000.00
7	Rohan Desai	BSc CS	Mumbai	69	42000.00
8	Meera Kulkarni	BCA	Nashik	82	40000.00
9	Arjun Nair	BSc IT	Thane	59	45000.00
10	Divya More	BCA	Mumbai	93	40000.00

Q2) Insert at least 4 records into the Course table and 6 records into the Student table. Make sure every student has a valid CourseID from the Course table.

-- Insert records into Course table (based on CourseID and Course values in your data)

```
INSERT INTO Course (CourseID, CourseName, Fees) VALUES
```

```
(101, 'BSc IT', 45000.00),
```

```
(102, 'BCA', 40000.00),
```

```
(103, 'BSc CS', 42000.00),
```

```
(104, 'BCom', 40000.00);
```

-- Insert records into Student table (using StudentID, StudentName, City, and CourseID from your data)

```
INSERT INTO Student (StudentID, StudentName, Age, City, CourseID) VALUES
```

```
(1, 'Rahul Sharma', 21, 'Mumbai', 101),
```

```
(2, 'Priya Patil', 22, 'Pune', 102),
```

```
(3, 'Aman Verma', 21, 'Nashik', 103),
```

```
(4, 'Sneha Joshi', 20, 'Thane', 102),
```

```
(5, 'Karan Mehta', 22, 'Thane', 102),
```

```
(6, 'Neha Shah', 21, 'Pune', 101),
```

```
(7, 'Rohan Desai', 21, 'Mumbai', 103),
```

```
(8, 'Meera Kulkarni', 20, 'Nashik', 102),
```

```
(9, 'Arjun Nair', 21, 'Thane', 101),
```

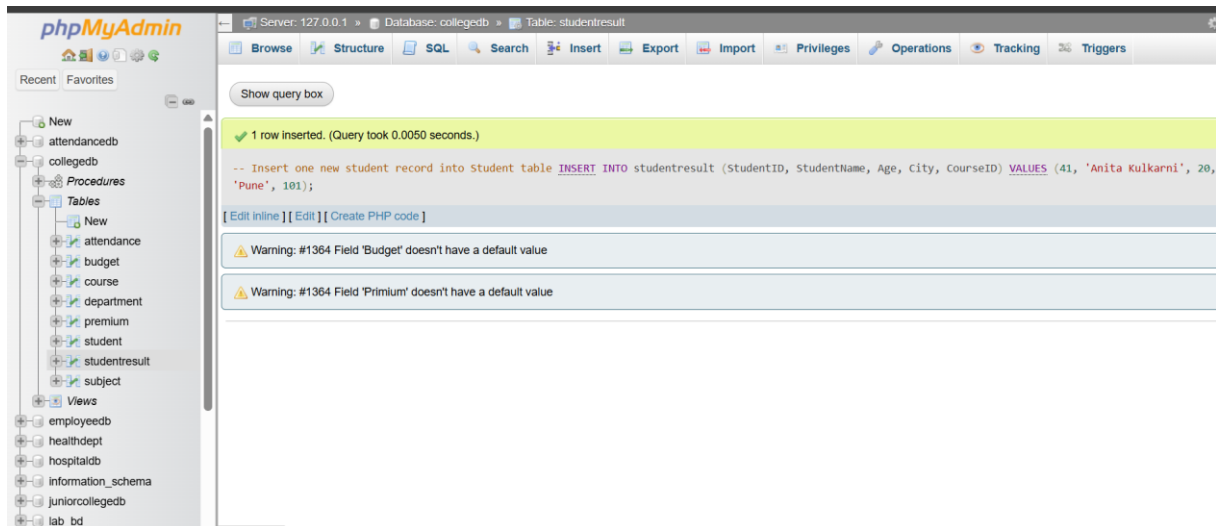
```
(10, 'Divya More', 22, 'Mumbai', 102);
```

Q3) Write a MySQL query to insert one new student record into the Student table.

The record should include: StudentID, StudentName, Age, City, CourseID.

-- Insert one new student record into Student table

```
INSERT INTO studentresult (StudentID, StudentName, Age, City, CourseID) VALUES  
(14, 'Anita Kulkarni', 20, 'Pune', 101);
```



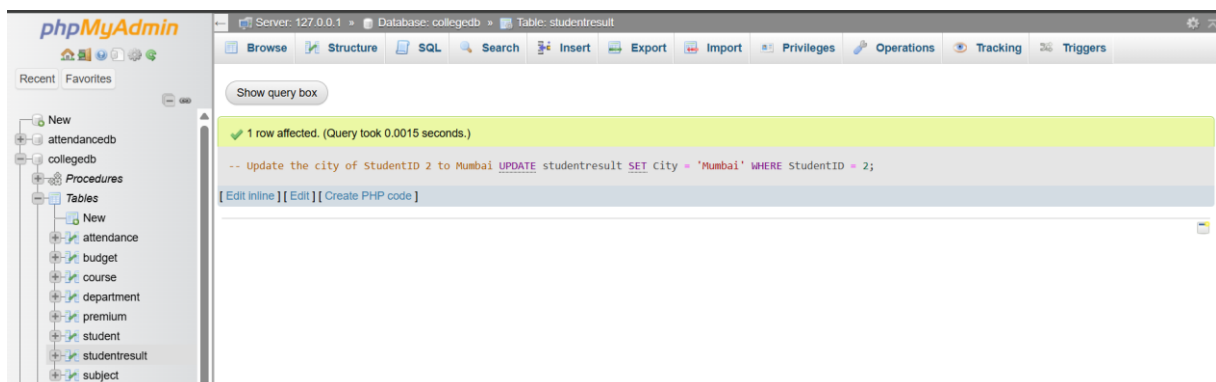
Q4) Write a MySQL query to update the city of a student using StudentID. (Example requirement: Update the city of StudentID 2 to Mumbai)

- Update the city of StudentID 2 to Mumbai

```
UPDATE studentresult
```

```
SET City = 'Mumbai'
```

```
WHERE StudentID = 2;
```



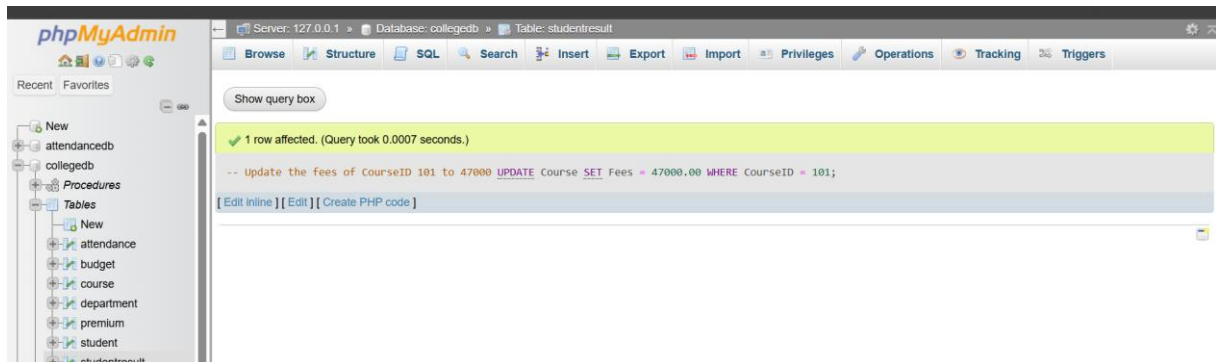
Q5) Write a MySQL query to update the fees of a course using CourseID. (Example requirement: Update the fees of CourseID 101 to 47000)

-- Update the fees of CourseID 101 to 47000

UPDATE Course

SET Fees = 47000.00

WHERE CourseID = 101;



The screenshot shows the 'studentresult' table data in phpMyAdmin. The table has 14 columns: StudentID, StudentName, Course, City, Marks, FeesPaid, CourseID, Budget, Premium, AdmissionDate, Phone, DateOfBirth, Gender, and Age. The data is as follows:

StudentID	StudentName	Course	City	Marks	FeesPaid	CourseID	Budget	Premium	AdmissionDate	Phone	DateOfBirth	Gender	Age
1	Rahul Sharma	BSc IT	Mumbai	78	45000.00	101	0	0	2026-06-10	9876543210	2005-01-15	M	21
2	Priya Patil	BCA	Mumbai	85	40000.00	102	0	0	2026-06-12	9765432109	2004-03-22	F	22
3	Aman Verma	BSc CS	Nashik	67	42000.00	101	0	0	2026-06-15	9812345678	2005-07-10	M	21
4	Sneha Joshi	BSc IT	Mumbai	91	45000.00	103	0	0	2026-06-18	9712345678	2006-02-18	F	20
5	Karan Mehta	BCA	Thane	74	40000.00	102	0	0	2026-06-20	9898765432	2005-09-25	M	20
6	Neha Shah	BSc IT	Pune	88	45000.00	101	0	0	2026-06-22	9787654321	2004-11-30	F	21
7	Rohan Desai	BSc CS	Mumbai	69	42000.00	104	0	0	2026-06-25	9823456789	2005-06-12	M	21
8	Meera Kulkarni	BCA	Nashik	82	40000.00	103	0	0	2026-06-27	9723456789	2006-04-05	F	20
9	Arjun Nair	BSc IT	Thane	59	45000.00	104	0	0	2026-06-29	9845678901	2005-08-19	M	20
10	Divya More	BCA	Mumbai	93	40000.00	102	0	0	2026-07-01	9745678901	2004-12-28	F	21
11	Sanjay Pawar	NULL	Nagpur	NULL	NULL	104	0	0	NULL	NULL	NULL	NULL	NULL
13	Sanjay Pawar	NULL	Nagpur	NULL	NULL	104	0	0	NULL	NULL	NULL	NULL	NULL
14	Anita Kulkarni	BSc IT	Pune	84	45000.00	101	0	0	2026-07-10	9876512345	2005-03-15	NULL	21
41	Anita Kulkarni	NULL	Pune	NULL	NULL	101	0	0	NULL	NULL	NULL	NULL	20

Q6) Write a MySQL query to delete one student record using StudentID. Also write one line explaining why the WHERE clause is important while using DELETE.

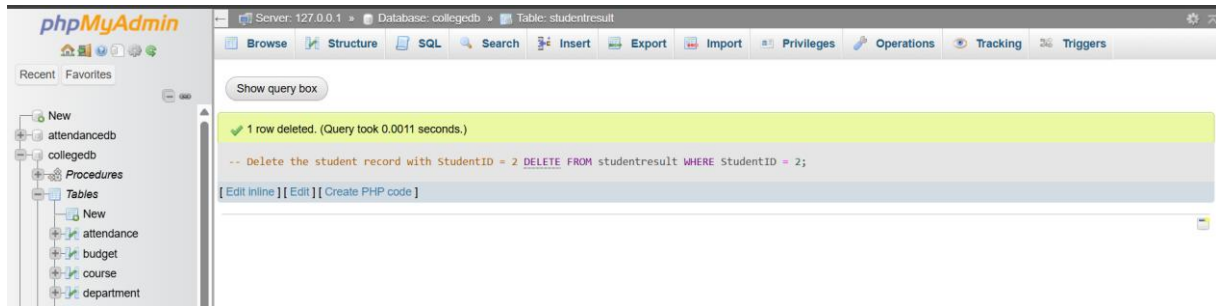
-- Delete the student record with StudentID = 2

DELETE FROM studentresult

WHERE StudentID = 2;

Why the WHERE clause is important

The WHERE clause is **critical** because without it, MySQL would delete **all rows in the table**, not just the one you intend to remove.



This screenshot shows the phpMyAdmin interface displaying the contents of the 'studentresult' table. A yellow status bar at the top indicates 'Showing rows 0 - 12 (13 total, Query took 0.0006 seconds.)'. The SQL query shown is `SELECT * FROM `studentresult``. Below the query, there are options for Profiling, Show all, Number of rows (set to 25), Filter rows, and Sort by key. The table data is presented in a grid with columns: StudentID, StudentName, Course, City, Marks, FeesPaid, CourseID, Budget, Premium, AdmissionDate, Phone, and DateOfBirth. Each row includes action links for Edit, Copy, and Delete.

	StudentID	StudentName	Course	City	Marks	FeesPaid	CourseID	Budget	Premium	AdmissionDate	Phone	DateOfBirth
☐	1	Rahul Sharma	BSc IT	Mumbai	78	45000.00	101	0	0	2026-06-10	9876543210	2005-01-15
☐	3	Aman Verma	BSc CS	Nashik	67	42000.00	101	0	0	2026-06-15	9812345678	2005-07-10
☐	4	Sneha Joshi	BSc IT	Mumbai	91	45000.00	103	0	0	2026-06-18	9712345678	2006-02-18
☐	5	Karan Mehta	BCA	Thane	74	40000.00	102	0	0	2026-06-20	9898765432	2005-09-25
☐	6	Neha Shah	BSc IT	Pune	88	45000.00	101	0	0	2026-06-22	9787654321	2004-11-30
☐	7	Rohan Desai	BSc CS	Mumbai	69	42000.00	104	0	0	2026-06-25	9823456789	2005-06-12
☐	8	Meera Kulkarni	BCA	Nashik	82	40000.00	103	0	0	2026-06-27	9723456789	2006-04-05
☐	9	Arjun Nair	BSc IT	Thane	59	45000.00	104	0	0	2026-06-29	9845678901	2005-08-19
☐	10	Divya More	BCA	Mumbai	93	40000.00	102	0	0	2026-07-01	9745678901	2004-12-28
☐	11	Sanjay Pawar	NULL	Nagpur	NULL	NULL	104	0	0	NULL	NULL	NULL
☐	13	Sanjay Pawar	NULL	Nagpur	NULL	NULL	104	0	0	NULL	NULL	NULL